

Charlie Wu

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Current Position

2026–present **Postdoctoral Fellow**, Max Planck Institute for Mathematics
Bonn, Germany

Education

2021–2026 **PhD in Mathematics**, University of Toronto
Toronto, Ontario
Advisor: Daniel Litt
Thesis: *Character Varieties and Hodge Theory*

Research Interests

Algebraic geometry and number theory.

Publications and Preprints

2025 **Hypergeometric Local Systems and Parabolic Bundles**
[arXiv:2509.25629](#)

2025 **Minimal Energy Local Systems on Curves**
Accepted at *Épijournal de Géométrie Algébrique*
[arXiv:2503.19250](#)

Talks

2026 *Privileged SL_3 -local systems* (joint talk with Julia Meng)
A Sprint on Local Systems, University of Wuppertal

2026 *Introduction to Higgs bundles*
Geometric Correspondences in the Desert, Anza-Borrego Desert State Park

2026 *Compact components of character varieties*
Algebraic Geometry Seminar, Johns Hopkins University

2026 *Compact components of character varieties*
Algebraic Geometry Seminar, University of Chicago

2025 *Compactness and character varieties*
Ottawa/Carleton Number Theory Seminar, University of Ottawa

2025 *Minimal energy local systems on curves*
Algebraic Geometry Seminar, University of Toronto

2024 *Non-abelian Hodge theory and complex variations of Hodge structure*
Elden Elmanto's Summer Seminar, University of Toronto

2024 *Six functor formalisms*
Étale Cohomology Seminar, University of Toronto

2024 *Intro to Higgs bundles and non-abelian Hodge theory*
Non-Abelian Hodge Theory Learning Seminar, University of Toronto

2024 *Étale fundamental groups*
Étale Cohomology Seminar, University of Toronto

2023	<i>Special unitary local systems on the Riemann sphere</i> Lightning Talk, Monodromy and Its Applications Conference (Nick Katz's 80th Birthday Conference), Princeton University
2023	<i>Why an engineer can draw better than you</i> Graduate Student Seminar, University of Toronto
2023	<i>Counting curves and Gromov-Witten invariants</i> Graduate Student Algebraic Geometry Seminar, University of Toronto
2023	<i>Character varieties</i> Rigid Local Systems Seminar, University of Toronto

Awards

2025	Margaret Isobel Elliott Graduate Scholarship
2018 & 2020	Drew Thompson Scholarships

Teaching Experience

University of Toronto

2026	MAT1301 — Algebraic Topology
2025	MAT138 — Intro to Proofs
2023	MAT224 — Linear Algebra II
2023	MAT237 — Multivariable Calculus with Proofs
2022–2025	MAT334 — Complex Variables
2022	MAT137 — Calculus with Proofs
2022	APM346 — Partial Differential Equations
2021	MAT344 — Introduction to Combinatorics
2019–2022	MAT223 — Linear Algebra I
2019	MAT136 — Calculus 1B
2018, 2021, 2022	MAT135 — Calculus 1A

Service

2024	Learning Seminar Organizer Non-Abelian Hodge Theory
2023–2024	Mentor to First-Year Graduate Students Advised on core courses, qualifying exams, finding an advisor, and related topics.
2020–2021	Vice-President of Social Affairs University of Toronto Mathematics Union

Citizenship: United States